

Discretizing a continuous bath: which frequency grid, and why

Notes for the Kac–Zwanzig test bed

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Abstract

A Kac–Zwanzig bath with N oscillators of frequencies ω_j and kernel weights k_j has the memory kernel $K_N(t) = \sum_j k_j \cos \omega_j t$, exactly. Choosing a grid (ω_j, k_j) to represent a continuous spectral density $J(\omega)$ is therefore nothing but choosing a quadrature rule for the cosine transform $K_\infty(t) = \frac{2}{\pi} \int_0^\infty \frac{J(\omega)}{\omega} \cos \omega t d\omega$. These notes work out the error $K_N - K_\infty$ for the two grids in use, the *midpoint-uniform* grid and the *inverse-CDF* grid, for the Ohmic and Debye spectral densities. The uniform grid’s error is a sum of smooth, alternating-sign, shifted copies of the kernel itself (Poisson summation), of size $2K_\infty(2\pi/\Delta\omega)$ in the working window, whereas the inverse-CDF grid leaves a high-frequency ripple from the under-resolved top of the spectrum. Which one wins depends on how smooth $J(\omega)/\omega$ is at $\omega = 0$ and how heavy its tail is: for Ohmic the uniform grid is better by three orders of magnitude at equal N ; for Debye the tan grid gets the cusp value $K(0)$ exactly but ripples at the 10^{-3} level, while a uniform grid with a generous cutoff is smoother everywhere except within $t \lesssim 1/\omega_{\max}$ of the cusp. In all cases the particle’s own correlation function converges faster than the kernel, because high-frequency bath modes couple to the particle with weight $\propto k_j/\omega_j^2$.

1 Setting

Bath masses are one; the model has two arrays, frequencies ω_j and kernel weights $k_j = c_j^2/(m_j\omega_j^2) \geq 0$. The particle feels the bath only through

$$K_N(t) = \sum_{j=1}^N k_j \cos \omega_j t, \quad K_\infty(t) = \int_0^\infty f(\omega) \cos \omega t d\omega, \quad f(\omega) \equiv \frac{2}{\pi} \frac{J(\omega)}{\omega}, \quad (1)$$

and $K_0 \equiv K_N(0) = \sum_j k_j$ approximates $\int_0^\infty f = K_\infty(0)$. Two facts frame everything below.

1. **K_N is exact for the finite bath.** The velocity autocorrelation function $C_N(t)$ of the particle coupled to the N oscillators satisfies $M\dot{C}_N = -\int_0^t K_N(t-s)C_N(s)ds$ identically, for every N and every grid. For testing a kernel-extraction method, K_N is the reference, and the grid question is only about how “continuum-like” the test problem looks over a window $[0, T]$.
2. **A grid is a quadrature rule.** Any choice of nodes ω_j and weights k_j is an approximation $\sum_j k_j g(\omega_j) \approx \int_0^\infty f(\omega)g(\omega)d\omega$ applied to $g(\omega) = \cos \omega t$. The integrand oscillates faster as t grows, so the accuracy of every grid degrades with t , and the question is always: how, and how fast.

The two spectral densities in use, in the Caldeira–Leggett convention $J(\omega) = \frac{\pi}{2} \sum_j k_j \omega_j \delta(\omega - \omega_j)$:

$$\text{Ohmic: } J = \eta \omega e^{-\omega/\omega_c}, \quad f(\omega) = \frac{2\eta}{\pi} e^{-\omega/\omega_c}, \quad K_\infty(t) = \frac{2\eta\omega_c/\pi}{1 + \omega_c^2 t^2}, \quad (2)$$

$$\text{Debye: } J = \frac{2\lambda\omega_c \omega}{\omega_c^2 + \omega^2}, \quad f(\omega) = \frac{4\lambda\omega_c/\pi}{\omega_c^2 + \omega^2}, \quad K_\infty(t) = 2\lambda e^{-\omega_c t}. \quad (3)$$

The essential difference: the Ohmic f is smooth for $\omega > 0$ but its *even extension* to the whole line, $f(|\omega|)$, has a kink at $\omega = 0$ ($f'(0^+) = -f(0)/\omega_c \neq 0$), and it decays exponentially. The Debye $f(|\omega|)$ is analytic at $\omega = 0$ but decays only like ω^{-2} : a heavy tail.

2 The midpoint-uniform grid

2.1 Definition

$$\omega_j = (j - \tfrac{1}{2})\Delta\omega, \quad k_j = f(\omega_j) \Delta\omega, \quad j = 1, \dots, N, \quad \omega_{\max} = N\Delta\omega. \quad (4)$$

This is the composite midpoint rule for $\int_0^{\omega_{\max}} f(\omega) \cos \omega t d\omega$. Its error has two pieces: truncation of $[\omega_{\max}, \infty)$, which we make negligible by choosing $\omega_{\max}$ (for Ohmic, $e^{-25} \approx 10^{-11}$), and the discretization error of the midpoint rule, which is the interesting part.

2.2 Poisson summation: the error is a sum of images

Write $g(\omega) = f(|\omega|) \cos \omega t$, even in ω . Then, ignoring truncation,

$$K_N(t) = \sum_{j=1}^N f(\omega_j) \Delta\omega \cos \omega_j t = \frac{\Delta\omega}{2} \sum_{j \in \mathbb{Z}} g((j - \tfrac{1}{2})\Delta\omega), \quad (5)$$

because the half-integer lattice is symmetric under $\omega \rightarrow -\omega$. The Poisson summation formula for a lattice offset by half a spacing reads

$$\Delta\omega \sum_{j \in \mathbb{Z}} g((j - \tfrac{1}{2})\Delta\omega) = \sum_{m \in \mathbb{Z}} (-1)^m \hat{g}\left(\frac{2\pi m}{\Delta\omega}\right), \quad \hat{g}(\tau) = \int_{-\infty}^{\infty} g(\omega) e^{-i\omega\tau} d\omega. \quad (6)$$

(The factor $(-1)^m = e^{i\pi m}$ comes from the half-spacing shift; without the shift it is absent.) With $g = f(|\omega|) \cos \omega t$ and $\hat{f}(\tau) \equiv \int_{-\infty}^{\infty} f(|\omega|) e^{-i\omega\tau} d\omega = 2K_\infty(\tau)$,

$$\hat{g}(\tau) = K_\infty(\tau - t) + K_\infty(\tau + t), \quad (7)$$

so that, using the evenness of K_∞ ,

$$\boxed{K_N(t) - K_\infty(t) = \sum_{m \neq 0} (-1)^m K_\infty\left(t - \frac{2\pi m}{\Delta\omega}\right)} = \sum_{m \geq 1} (-1)^m \left[K_\infty(t - mP) + K_\infty(t + mP) \right], \quad (8)$$

with $P \equiv 2\pi/\Delta\omega$ the period of the lattice sum. Figure 1 checks this identity numerically: for $N = 400$, $\Delta\omega = 0.0625$, the computed $K_N - K_\infty$ and the image sum agree to 2×10^{-9} , which is the truncation level of the sum.

Everything about the uniform grid follows from (8):

- **The error is smooth.** It is a sum of shifted copies of K_∞ itself, not a ripple. In the working window $t \ll P$ the $m = \pm 1$ images give $K_N - K_\infty \approx -2K_\infty(P)$, a slowly varying negative offset. For the Ohmic Lorentzian, summing the alternating series, $K_N - K_\infty \approx -2K_0 \sum_{m \geq 1} (-1)^{m+1} / (1 + (m\omega_c P)^2) \approx -\frac{K_0}{2\pi^2} \left(\frac{\Delta\omega}{\omega_c}\right)^2 \times 0.82$; for $N = 2000$, $\omega_{\max} = 25\omega_c$ this is -4.3×10^{-6} , exactly the measured value (Fig. 2, top).
- **Periodicity and window.** K_N is exactly periodic with period $P = 2\pi/\Delta\omega$ (all ω_j are odd multiples of $\Delta\omega/2$) and even about $P/2$. The first image, at $t = P$, is a *negative* copy of the kernel (Fig. 2, bottom). The usable window is $T < P/2 = \pi/\Delta\omega$; for $N = 2000$, $\omega_{\max} = 25$ that is $T < 251$.
- **Order of accuracy is set by the smoothness of $f(|\omega|)$ at $\omega = 0$.** The images decay like the Fourier transform of the even extension. For Ohmic, the kink at 0 makes $K_\infty(\tau) \sim \tau^{-2}$, so the error is $O(\Delta\omega^2)$: the midpoint rule is *second order* here, not spectral. It agrees with the Euler–Maclaurin leading term $-\frac{\Delta\omega^2}{24} f'(0^+)$ at $t = 0$. The measured N -scan (Fig. 2, middle) follows $2K_\infty(P) \propto N^{-2}$ for $N \geq 1000$; the two smallest N lie above the line because there $P/2 < 50$ and the first image sits inside the window. For Debye, $f(|\omega|)$ is analytic, $K_\infty(\tau) \propto e^{-\omega_c|\tau|}$, and the images are exponentially small ($e^{-\omega_c P}$): the midpoint rule is spectral for the smooth part; only the tail truncation remains (Section 4).
- **Midpoints, not endpoints.** The grid $\omega_j = j\Delta\omega$ is the same lattice without the half shift. Poisson summation then has no $(-1)^m$, and the sum $\sum_{j \geq 1} = \frac{1}{2}[\sum_{j \in \mathbb{Z}} - g(0)]$ leaves behind

$$K_N^{\text{end}}(t) - K_\infty(t) = -\frac{1}{2}\Delta\omega f(0) + \sum_{m \neq 0} K_\infty(t - mP) = -\frac{\Delta\omega \eta}{\pi} + O(\Delta\omega^2), \quad (9)$$

a constant bias (4×10^{-3} at $N = 2000$; measured 3.97×10^{-3}), a thousand times the midpoint error and first order in $\Delta\omega$. It is the trapezoid rule missing its half weight at the origin.

3 The inverse-CDF grid

3.1 Definition

Let $\rho(\omega) = f(\omega)/K_0$, a probability density, and let $u(\omega) = \int_\omega^\infty \rho$ be its upper tail. The inverse-CDF grid places nodes at the quantiles and gives them equal weights:

$$u_j = \frac{j - \frac{1}{2}}{N}, \quad \omega_j = u^{-1}(u_j), \quad k_j = \frac{K_0}{N}. \quad (10)$$

For Ohmic, $u = e^{-\omega/\omega_c}$ and $\omega_j = -\omega_c \ln \frac{j-1/2}{N}$, the grid of the “Discrete Oscillators” notebook. For Debye, $u = 1 - \frac{2}{\pi} \arctan(\omega/\omega_c)$ and $\omega_j = \omega_c \tan\left(\frac{\pi}{2} \frac{j-1/2}{N}\right)$, the notebook’s tan grid. Two properties are immediate: K_0 is reproduced exactly (equal weights sum to it), and there is no truncation, since the quantiles cover the whole support (the largest node is $\omega_N = \omega_c \ln(2N)$ for Ohmic, $\approx \frac{4N}{\pi} \omega_c$ for Debye).

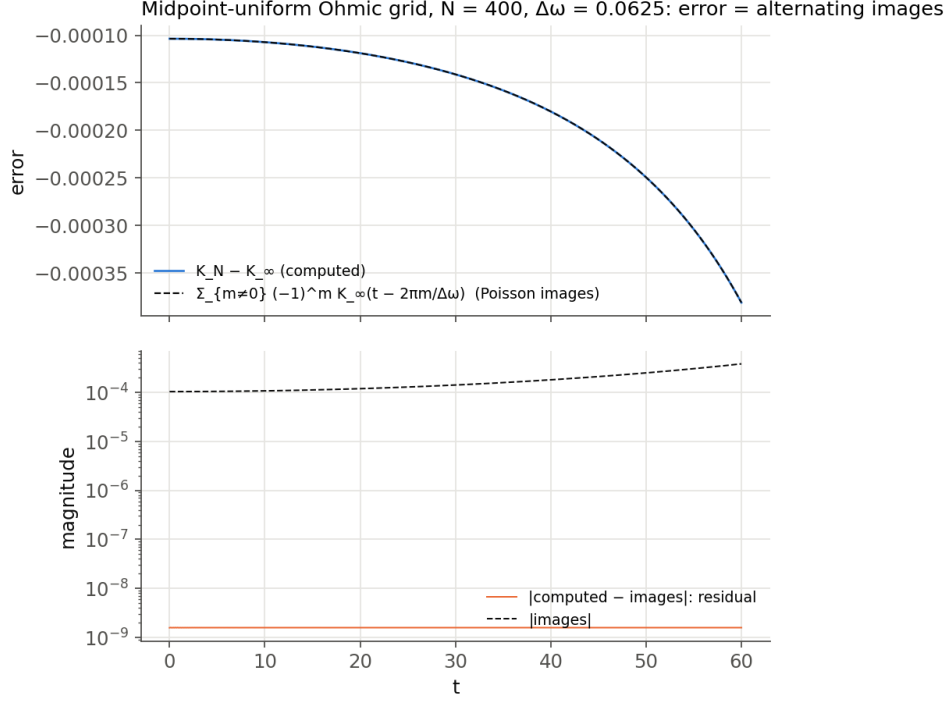


Figure 1: The Poisson-image identity (8) for the midpoint-uniform Ohmic grid, $N = 400$, $\Delta\omega = 0.0625$ ($P = 100.5$): the computed error (solid) and the alternating image sum (dashed) coincide; the residual is at the 10^{-9} truncation level of the image sum.

3.2 It is the midpoint rule in the cumulative variable

Changing variables in (1),

$$K_\infty(t) = K_0 \int_0^1 \cos(\omega(u)t) du, \quad K_N(t) = \frac{K_0}{N} \sum_{j=1}^N \cos(\omega(u_j)t), \quad (11)$$

so the inverse-CDF grid is the composite midpoint rule, spacing $\Delta u = 1/N$, for the integrand $h(u) = \cos(\omega(u)t)$. The catch is at $u \rightarrow 0$, where $\omega(u) \rightarrow \infty$ and h oscillates infinitely fast: $h'(u) = -t\omega'(u)\sin(\omega t)$ with $\omega'(u) = -\omega_c/u$ (Ohmic), so $|h'| \sim \omega_c t/u$ is unbounded and the Euler–Maclaurin expansion does not exist. The rule is accurate only where the phase advance between neighbouring nodes is small,

$$\delta\phi_j = t|\omega'(u_j)|\Delta u = \frac{\omega_c t}{j - \frac{1}{2}} \ll 1 \quad \Longleftrightarrow \quad j \gg \omega_c t. \quad (12)$$

The first $n_{\text{scr}} \approx \omega_c t$ nodes, the highest frequencies, are *under-resolved*: their cosines have effectively random phases and do not cancel the way the integral does. Adding n_{scr} terms of size K_0/N with random signs gives

$$|K_N - K_\infty| \approx \frac{K_0}{N} \sqrt{\omega_c t} \quad (\text{Ohmic, inverse-CDF}), \quad (13)$$

a *ripple* whose frequencies are those of the scrambled modes, $\omega \sim \omega_c \ln(N/(\omega_c t))$, i.e. high; whose amplitude grows like $\sqrt{t}$; and which decays only like N^{-1} . At $t = 50$, $N = 2000$: $0.64 \times 7.1/2000 =$

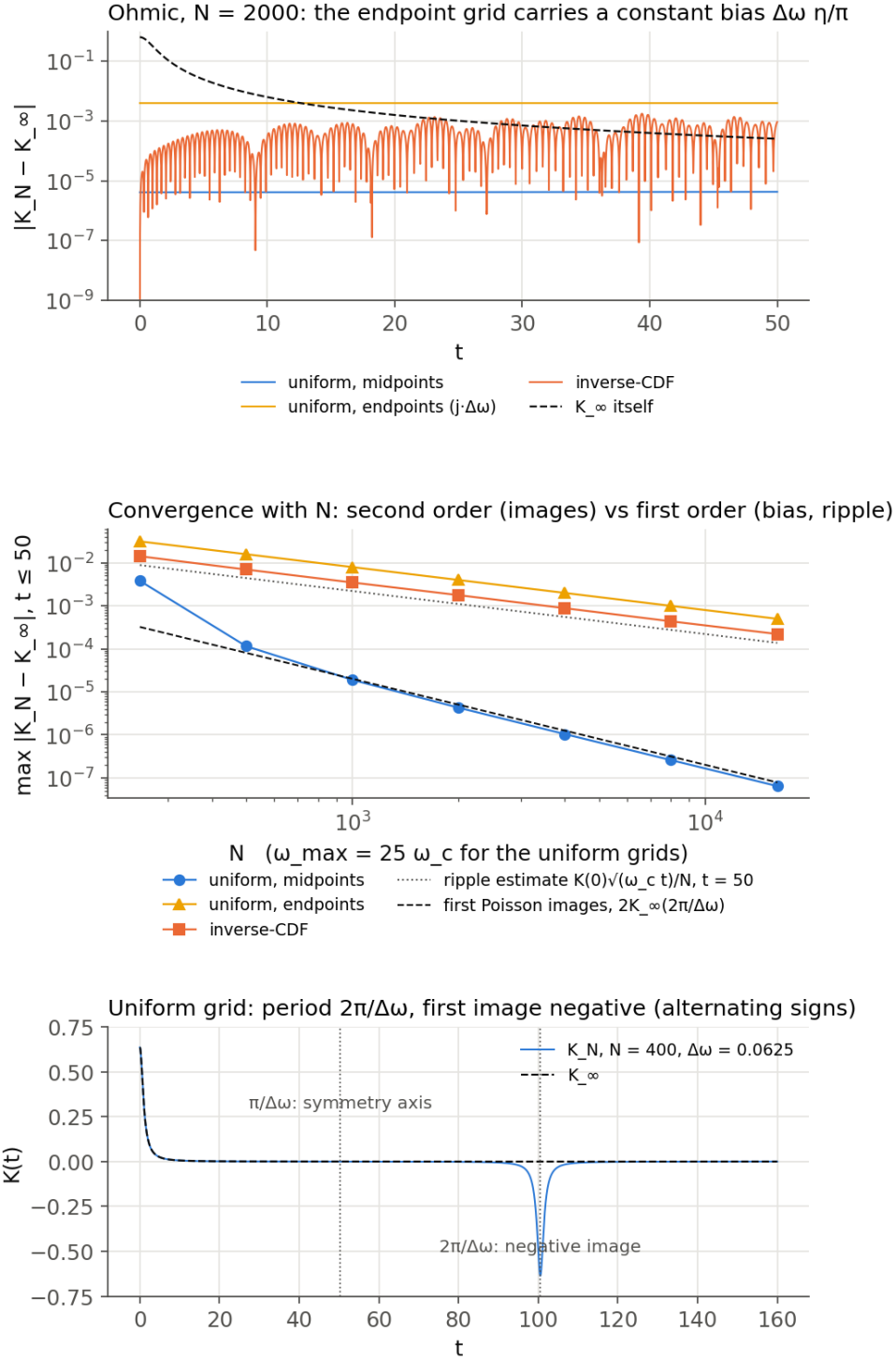


Figure 2: Ohmic bath, $\eta = \omega_c = 1$. Top: $|K_N - K_\infty|$ at $N = 2000$ for the midpoint-uniform grid (smooth offset 4×10^{-6}), the endpoint grid (constant bias $\Delta\omega \eta/\pi$) and the inverse-CDF grid (ripple, growing like $\sqrt{t}$). Middle: the maximum error over $t \leq 50$ against N ; the midpoint grid follows the first-image prediction $2K_\infty(2\pi/\Delta\omega) \propto N^{-2}$, the other two are $\propto N^{-1}$. Bottom: periodicity of the uniform grid.

$\max_{t \leq 50} K_N - K_\infty $, Ohmic, $K_0 = 0.64$	$N = 2000$	$N = 20000$	$N = 10^5$
midpoint-uniform, $\omega_{\max} = 25\omega_c$	4×10^{-6}	4×10^{-8}	2×10^{-9}
inverse-CDF (notebook)	2×10^{-3}	2×10^{-4}	4×10^{-5}

Table 1: Measured kernel errors for the Ohmic bath. The continuum tail is $K_\infty(50) = 2.5 \times 10^{-4}$, so the inverse-CDF ripple at $N = 2000$ is larger than the kernel it is supposed to follow there.

2.3×10^{-3} , against the measured 1.8×10^{-3} (Fig. 2). Both the N^{-1} scaling and the $\sqrt{t}$ growth are visible in the figure.

For the Ohmic bath this settles the comparison: at $N = 2000$ the uniform grid is closer to the continuum by a factor 400, and at $N = 10^5$ the inverse-CDF grid (3.5×10^{-5}) is still eight times worse than the uniform grid at $N = 2000$. Table 1 collects the measured values. The inverse-CDF grid is not wrong, and it is the natural choice when what matters is the total weight and the low-frequency end (it puts nodes down to $\omega_1 \approx \omega_c/2N$ and never truncates), but it is the wrong tool for a smooth kernel over a long window.

4 The Debye bath: heavy tail

Here $f(\omega) \propto (\omega_c^2 + \omega^2)^{-1}$, so half of the weight above ω scales like ω_c/ω : there is no natural cutoff.

Uniform grid with a cutoff. Two error sources. (i) The truncated tail,

$$K_\infty(t) - \int_0^{\omega_{\max}} f \cos \omega t d\omega = \frac{4\lambda\omega_c}{\pi} \int_{\omega_{\max}}^{\infty} \frac{\cos \omega t}{\omega_c^2 + \omega^2} d\omega = \begin{cases} \frac{2}{\pi} K_0 \arctan \frac{\omega_c}{\omega_{\max}}, & t = 0, \\ O\left(\frac{K_0 \omega_c}{\omega_{\max}^2 t}\right), & t \gg 1/\omega_{\max}, \end{cases} \quad (14)$$

(integration by parts kills the $1/\omega_{\max}$ term as soon as $\omega_{\max}t \gg 1$). So the cutoff costs a deficit of 0.8% ($\omega_{\max} = 100\omega_c$) or 0.2% ($400\omega_c$) of K_0 right at the cusp, which disappears within $t \lesssim 1/\omega_{\max}$, after which the truncation error is 10^{-5} or smaller. (ii) The Poisson images: $f(|\omega|)$ is analytic, so they are $O(e^{-\omega_c P})$, invisible. The price of a large cutoff at fixed N is a coarser $\Delta\omega$ and a shorter window $\pi/\Delta\omega$ (31 time units for $\Delta\omega = 0.1$).

The tan grid. Same construction as before, now with $\omega(u) = \omega_c \tan(\pi u/2)$ and $\omega'(u) = \frac{\pi\omega_c}{2}(1 + \omega^2/\omega_c^2)$. The phase advance per node exceeds one for $\omega \gtrsim \omega_c \sqrt{2N/(\pi\omega_c t)}$; the fraction of nodes above that frequency is $\approx \frac{2}{\pi} \sqrt{\pi\omega_c t/(2N)}$, so the number of scrambled nodes is $n_{\text{scr}} \propto \sqrt{Nt}$ rather than $\propto t$, and the ripple estimate becomes

$$|K_N - K_\infty| \approx \frac{K_0}{N} \sqrt{n_{\text{scr}}} \propto K_0 t^{1/4} N^{-3/4} \quad (\text{Debye, tan grid}). \quad (15)$$

The measured convergence exponent is -0.77 (errors 5.7×10^{-2} , 1.9×10^{-2} , 6.7×10^{-3} at $N = 250, 1000, 4000$). Because the tail is heavy, far more weight sits in scrambled modes than for Ohmic, and the ripple at $N = 4000$ is a few 10^{-3} at all t , three orders of magnitude above the uniform grid with $\omega_{\max} = 400$ for $t > 1/\omega_{\max}$ (Fig. 3, top). What the tan grid does get right is $K(0) = 2\lambda$ exactly, and the cusp region $t \lesssim 1/\omega_{\max}$.

Why the particle does not care as much. The velocity autocorrelation function converges much faster than the kernel on every grid (Fig. 3, middle; tan grid exponent -1.7 , values 3.6×10^{-4} , 2.7×10^{-5} , 2.9×10^{-6}). The reason is in the particle's spectral measure. A bath mode of weight k_j at a frequency ω_j far above the particle's response band produces a normal mode with particle weight

$$a_k^2 \approx \frac{k_j}{M\omega_j^2} \quad (k_j \ll M\omega_j^2), \quad (16)$$

the familiar $1/(M\omega^2)$ of a free mass driven at high frequency. The scrambled modes are precisely the high-frequency ones, so their ripple enters C_N suppressed by $1/\omega_{\text{scr}}^2 \propto t/N$, turning $N^{-3/4}$ into $N^{-7/4}$ for the tan grid, as observed. The same mechanism suppresses the inverse-CDF ripple in the Ohmic case. For the kernel-extraction test this means: a noisy-looking K_N is not a symptom of a bad C_N , and a good C_N does not imply a smooth K_N ; the two converge at different rates, and the inversion must recover the ripple because it is real.

5 Summary and recommendations

	midpoint-uniform	inverse-CDF (equal weights)
what it is	midpoint rule in ω	midpoint rule in the cumulative variable u
error type	smooth alternating images of K_∞ at $t \pm 2\pi m/\Delta\omega$	high-frequency ripple from the under-resolved top of the spectrum
size in window	$\approx 2K_\infty(2\pi/\Delta\omega)$	$\approx (K_0/N)\sqrt{n_{\text{scr}}}$, $n_{\text{scr}} \sim \omega_c t$ (Ohmic), $\sim \sqrt{Nt}$ (Debye)
order in N	N^{-2} if $f(\omega)$ has a kink at 0 (Ohmic); spectral if analytic (Debye)	N^{-1} (Ohmic), $N^{-3/4}$ (Debye)
$K(0)$	off by $O(\Delta\omega^2)$, plus the truncated tail	exact
window	$\pi/\Delta\omega$, then an exact negative image	no exact period; quasi-recurrence $\sim 2\pi/\Delta\omega_{\text{min}}$, far away

- **Ohmic (exponential cutoff): midpoint-uniform.** $N = 2000$, $\omega_{\text{max}} = 25\omega_c$ gives $|K_N - K_\infty| \leq 4 \times 10^{-6}$ over $t \leq 250$, with a smooth error. Use midpoints; the endpoint grid is a thousand times worse. Do not expect spectral accuracy: the kink of $e^{-|\omega|/\omega_c}$ at zero makes it second order, but the constant is tiny.
- **Debye (Lorentzian J/ω): it depends on what you want.** If the cusp value $K(0) = 2\lambda$ and the first $1/\omega_{\text{max}}$ of time matter, use the tan grid and accept a 10^{-3} ripple at $N = 4000$ (decaying as $N^{-3/4}$). If the kernel's shape over the window matters more, a uniform grid with $\omega_{\text{max}} \gtrsim 400\omega_c$ is smoother by three orders of magnitude for $t > 1/\omega_{\text{max}}$; watch the window $\pi/\Delta\omega$. Either way C_N is excellent: 10^{-6} (tan) to 10^{-9} (uniform) at $N = 4000$.
- **The general rule.** Look at $f(|\omega|) = \frac{2}{\pi}J(|\omega|)/|\omega|$ on the whole line. Its smoothness at $\omega = 0$ sets the order of the uniform grid; its tail sets whether truncation is affordable. The

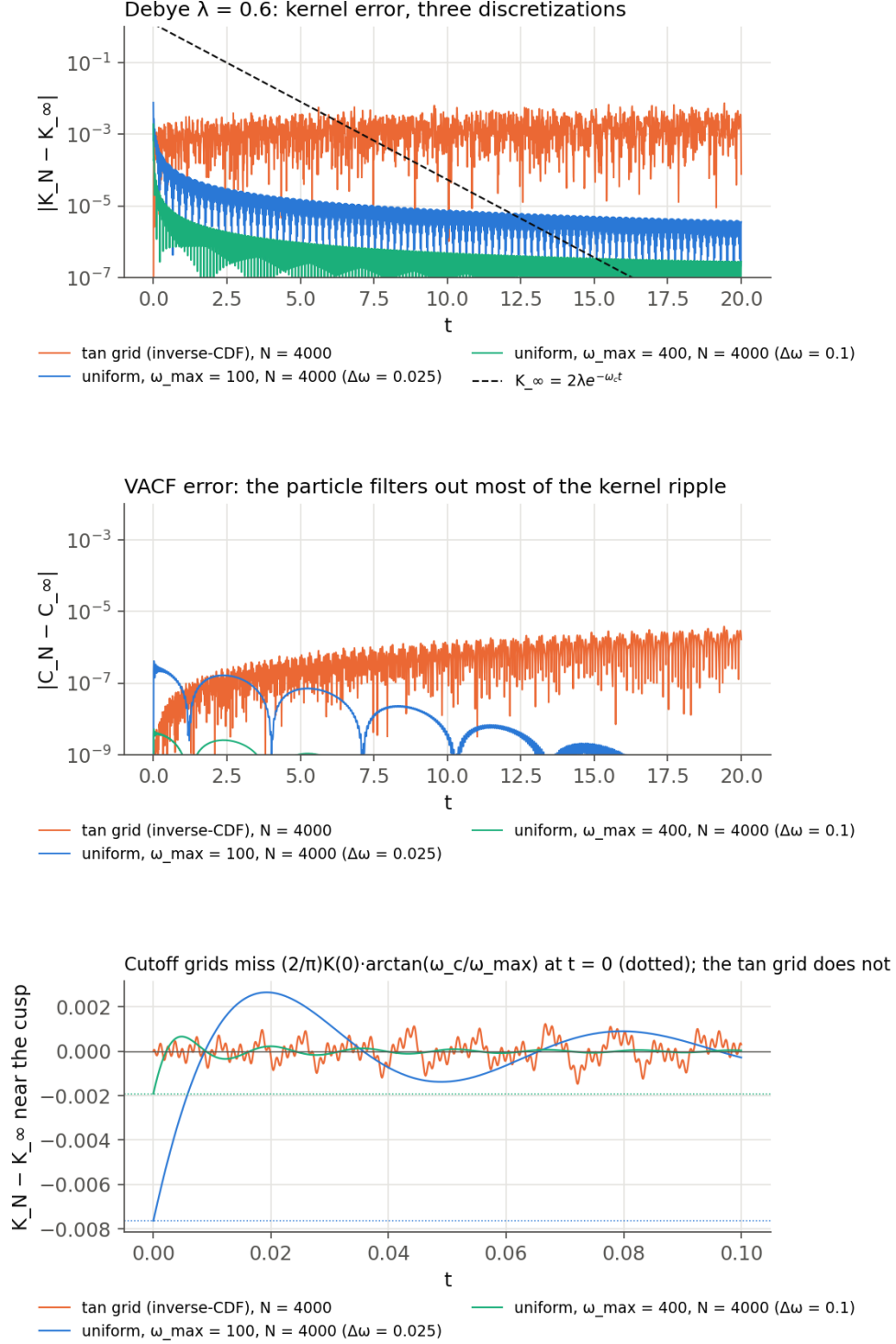


Figure 3: Debye bath, $\lambda = 0.6$, $\omega_c = 1$, $N = 4000$ on three grids. Top: kernel error; the tan grid ripples at 10^{-3} , the cutoff grids are smooth and small beyond $t \sim 1/\omega_{\max}$. Middle: VACF error, three to four orders smaller in every case. Bottom: the cusp region, where the cutoff grids miss $\frac{2}{\pi}K_0 \arctan(\omega_c/\omega_{\max})$ (dotted) and the tan grid does not.

inverse-CDF grid trades truncation for a ripple whose size is governed by how many of its equal-weight nodes sit above the frequency where the phase advance per node reaches one. When f decays exponentially, that number is $\sim \omega_c t$ and the ripple loses; when f decays algebraically, the uniform grid needs a large cutoff and the tan grid's exactness at $t = 0$ may be worth its ripple.

- **None of this changes the test.** K_N is the exact kernel of the finite bath and the reference for the inversion. The grid only decides how much the finite-bath kernel resembles the textbook continuum one over the window, and whether the reference is smooth or rippled.